

REVIEW ARTICLE

Immunotherapy for resectable lung cancer

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Abstract

Lung cancer remains a significant global health challenge, demanding innovative treatment strategies. Immune checkpoint blockade has revolutionized cancer care, leading to improved survival across advanced malignancies and has now become a standard therapy for earlier stage, resectable lung cancer. This review article consolidates the current landscape and future prospects of neoadjuvant and perioperative immunotherapy in lung cancer. The authors outline key findings from clinical trials in resectable lung cancer, including early efficacy, safety profiles, and emerging impact on disease recurrence, and overall survival. Additionally, this review elucidates the challenges encountered, including patient selection criteria, optimal treatment schedules, immune-related adverse events, and impact on surgery. This comprehensive analysis amalgamates current evidence with future directions, providing a roadmap for clinicians, researchers, and stakeholders to navigate the dynamic realm of immunotherapy for surgically resectable lung cancer.

KEYWORDS

adjuvant, immune checkpoint, immunotherapy, localized, lung cancer, neoadjuvant, non-small cell lung cancer (NSCLC), perioperative, resectable, surgery

INTRODUCTION

Lung cancer is associated with significant morbidity and mortality, with more than 2 million cases diagnosed worldwide, contributing to almost 1.8 million deaths annually.¹ Non-small cell lung cancer (NSCLC) is the most common subtype representing approximately 85% of cases, of which adenocarcinoma remains the most common histology. By 2050, the incidence of lung cancer is projected to increase to 3.8 million with an estimated 3.2 million deaths each year.² Historically, approximately one-third of lung cancers were traditionally considered surgically resectable (stage I–IIIA) at diagnosis. For these patients, 5-year survival rates remained between 38% to 60% despite incremental improvements in surgical approaches and optimization of perioperative management.^{3,4} Survival

after potentially curative surgery for lung cancer is lower than most other common cancers likely as a result of micrometastatic disease that is refractory to either perioperative chemotherapy or radiotherapy. This paradigm is shifting, given the contemporary results and inclusion of more locally advanced, multi-stage N2 disease (stage IIIB) in recent immunotherapy-based phase 3 trials. Earlier detection of lung cancer may increase over the coming decades with the expansion of CT screening programs and novel liquid biopsy assays. However, fewer than 17% of eligible patients at high risk for lung cancer currently undergo recommended computed tomography (CT) screening in the United States where it is available and even fewer are screened worldwide, despite projections that thousands of lives could be saved through effective implementation.⁵

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Immunotherapy targeting adaptive immunity and exhausted or depleted intra-tumoral T cells has radically changed the treatment landscape of multiple solid tumors, including advanced non-oncogene driven NSCLC, however, there is a clear need to improve therapeutic options for earlier stage disease. Positive results with the use of immune checkpoint inhibitors (ICI) as monotherapy (programmed cell death protein 1 and its ligand [PD-1/PD-L1] inhibitors), dual immune checkpoint blockade (anti-PD-1 and cytotoxic T-lymphocyte-associated protein 4 [CTLA-4]), or combined with chemotherapy or as consolidation after chemoradiation in locally advanced unresectable NSCLC cohorts have spurred investigation in potentially resectable NSCLC.^{6–11} PD-L1 tumor proportion score (TPS) expression acts as a predictive biomarker of response in the advanced lung cancer and is pivotal to clinicians in guiding drug selection, however, its utility in resectable lung cancer remains uncertain.¹²

The objective of this review article is to give a comprehensive overview of the current clinical practice paradigms and discuss the rationale for a neoadjuvant or perioperative approach in NSCLC. We will examine the evidence to date and critically analyze clinical data supporting the use of neoadjuvant, adjuvant, or perioperative ICI treatment, highlighting the respective merits and potential drawbacks. We will also explore novel biomarkers, including pathologic response and circulating tumor DNA (ctDNA) clearance that may be incorporated into future treatment algorithms for potentially resectable NSCLC.

ADJUVANT CHEMOTHERAPY AND IMMUNOTHERAPY

Adjuvant chemotherapy, consisting of four cycles of postoperative platinum doublet chemotherapy, has been the standard of care for stage IIA–IIIB resected NSCLC for over 15 years. The use of adjuvant chemotherapy is supported by the LACE meta-analysis of large phase 3 trials that showed a modest advantage in overall survival (4% to 5.4%) over surgery alone at 5 years.¹³ Neoadjuvant chemotherapy delivers a similar advantage and toxicity profile over surgery alone, albeit based on mostly smaller studies, some of which were not fully enrolled.¹⁴

The PD-L1 inhibitor atezolizumab was the first ICI approved by the Food and Drug Administration (FDA) as adjuvant therapy for NSCLC based on the landmark IMpower010 trial results for patients whose tumors expressed PD-L1 TPS $\geq 1\%$ of tumor cells.¹⁵ This study was the first to demonstrate the benefit of adjuvant immunotherapy after adjuvant chemotherapy. Patients with completely resected disease received adjuvant atezolizumab for 1 year or best supportive care (BSC), after adjuvant cisplatin-based chemotherapy. The primary end point was disease-free survival (DFS) hierarchically tested in the PD-L1 positive ($\geq 1\%$) population, all stage II–IIIA population, and finally the intention-to-treat (ITT) population. Atezolizumab improved DFS in the PD-L1 positive subpopulation of patients (hazard ratio [HR], 0.66; 95% confidence interval [CI], 0.50–0.88;

$p = .0039$), however, this benefit was less pronounced in the overall stage II–IIIA population (HR, 0.79; 95% CI, 0.64–0.96; $p = .02$). A post hoc exploratory analysis confirmed that the magnitude of benefit was closely correlated to the level of PD-L1 expression with the PD-L1 TPS $\geq 50\%$ population deriving the most benefit (PD-L1 $< 1\%$: HR, 0.97, 0.72–1.31; PD-L1 1%–49%: HR, 0.87, 0.60–1.26; and PD-L1 $\geq 50\%$: HR, 0.43, 0.27–0.66).¹⁵ Updated DFS and overall survival data from this trial was recently reported, and although still immature, the subgroup data shows an emerging survival benefit for patients with tumor PD-L1 $\geq 50\%$, however, no discernible survival benefit for other groups thus far.¹⁶ In light of this, the European Medicines Agency (EMA) Committee for Medicinal Products for Human Use has restricted the use of adjuvant atezolizumab to the subset of patients with tumor PD-L1 expression $\geq 50\%$. Treatment-related adverse events (TRAEs) were as reported previously with PD-1 pathway blockade, with 22% grade 3 or 4 adverse events and 8% of patients developing grade 3–4 immune-related adverse events (irAEs). irAEs led to treatment discontinuation in 11% of cases. A notable limitation in this trial is that just over half of patients were treated with PD-1 pathway blockade at the time of relapse, considered the standard of care in most settings for non-oncogene driven tumors. The utility of adjuvant immunotherapy has been further reinforced by the initial positive results of the recently published interim analysis of the PEARLS/KEYNOTE-091 study.¹⁷ Patients with completely resected disease were randomized to adjuvant pembrolizumab for 1 year or BSC, and although adjuvant chemotherapy was not mandatory was strongly recommended. Adjuvant pembrolizumab led to a significantly improved median DFS of 53.6 months (95% CI, 39.2–not reached) versus 42.0 months (95% CI, 31.3–not reached) in the placebo group (HR, 0.76; 95% CI, 0.63–0.91; $p = .014$). The study's dual primary end points were DFS in the overall population and in the PD-L1 TPS $\geq 50\%$ population. In a surprising finding, DFS was not yet significantly improved in the PD-L1 TPS $\geq 50\%$ population (HR, 0.82; 95% CI, 0.57–1.18; $p = .14$). In this study, 34% of patients developed grade 3 or worse adverse events with 20% of these events leading to treatment discontinuation.¹⁷ Most strikingly, adverse events leading to death were seen in 11 (2%) participants treated with pembrolizumab, compared to four patients (1%) in IMpower010.

SCIENTIFIC RATIONALE AND BIOLOGICAL BENEFITS OF NEOADJUVANT THERAPY

The purpose of any neoadjuvant treatment is to downstage patients before surgical resection and improve optimal resection rates (R0) and survival outcomes overall. Pathological regression in response to neoadjuvant therapy is an early indicator of therapeutic efficacy. Standard measurements are pathological complete response (pCR), defined as no viable tumor within the resected specimen, or major pathological response (MPR), defined as 10% or less of residual viable tumor cells in the resected primary tumor or lymph nodes sampled.

Other metrics used primarily in gastrointestinal cancers include Becker and Mandard's tumor regression grades 1–3 and 1–5, respectively.¹⁸ pCR after neoadjuvant treatment is a potential efficacy end point across solid tumors, such as early-stage breast cancer, melanoma, and microsatellite instability high rectal cancer.^{19–21} For NSCLC, the efficacy outcomes of MPR and pCR seen at surgical resection following neoadjuvant systemic chemotherapy have been shown to be positively correlated with improved DFS and overall survival (OS).^{22,23} There are physiological and cellular macroscopic and microscopic advantages of administering treatment in the neoadjuvant setting: intact blood vessels allow improved drug delivery to the target organ and tumor microenvironment (TME). With the presence of intact draining regional lymph nodes, there is the potential to enhance priming of the anti-tumor T-cell response toward both primary and micrometastatic tumor cells.

Delivery of immunotherapy in the neoadjuvant setting is hypothesized to generate an optimal immune response, leading to improved pathological responses and survival outcomes, DFS, and OS. The scientific rationale behind this combinatorial effect is that immune-complementary chemotherapy or radiotherapy act synergistically with immunotherapy as they stimulate endogenous immunogenic cell death with subsequent release of tumor-associated neoantigens promoting maturation of antigen-presenting cells (APC) in TME producing more polyclonal T-cells and priming responsiveness to immunotherapy to optimize therapeutic response.^{24,25} Another mechanism of synergy includes the upregulation of PD-L1 expression on the tumor cell surface after exposure to chemotherapy thus improving antitumor activity.²⁶ Unlike adjuvant therapy, the utilization of immunotherapy in the neoadjuvant setting enables objective *in vivo/ex vivo* observations of the tumor microenvironment and related immune editing across various immune compartments and at different time points.

Anti-CTLA-4 therapy, such as tremelimumab or ipilimumab, increase proliferation and differentiation of T cells whereas anti-PD-1 or PD-L1 therapies, including nivolumab, allow improved T-cell detection of tumor cells.²⁷ Although these therapies use different mechanisms of action, they act synergistically to activate the immune system and lead to increased tumoral detection. Dual CTLA-4 and PD-(L)1 inhibitor has been shown to be more effective than monotherapy in the pre-clinical setting, as well as improving overall response rates (ORR) and progression-free survival (PFS) in the clinical setting.^{27–34} However, this comes with the caveat of a potentially increased side effect profile. The synergistic effects of chemotherapy and immunotherapy to bolster the antitumoral effect of the immune response are well-documented.^{35,36} The immunomodulatory effects of chemotherapy enhance immunotherapy efficacy via multiple mechanisms: blockade of signal transducer and activator of transcription 6 (STAT6) signaling, the downregulation of PD-L1 and induction of mannose-6-phosphate receptor expression, and with stimulation of production and activation of high-mobility group protein box-1 to promote cell death of immunogenic tumor cells.^{37,38} Gemcitabine, for example, often used in NSCLC squamous

cell carcinoma subtype in combination with platinum (carboplatin or cisplatin), not only facilitates tumor infiltration with antigen-specific CD8⁺ T cells but also reduces the number of immunosuppressive cells, thus facilitating the eradication of tumors.³⁹ These effects can potentially convert immunologically cold tumors (tumors with no detectable inflammatory process or tumor infiltrating lymphocytes activity) to hot tumors with significant inflammation. The clinical benefits from a surgical perspective of combined neoadjuvant chemioimmunotherapies include the potential to decrease the size of tumors involved and ultimately downstage the primary tumor and lymph node burden and increase the possibility of complete surgical resection (R0) for patients with a large burden of disease (T4) and disease in anatomically inaccessible locations. Conversely, the major theoretical concerns regarding neoadjuvant immunotherapy and chemioimmunotherapy are that they may cause increased fibrosis and adhesions related to tumoral response, increasing the complexity associated with resection. In addition, there may be delays in surgical resection due to toxicities associated with neoadjuvant treatment, which may lead to worse DFS and OS for this subset of patients. Interim imaging is an essential part of the cadence of activities before surgical resection to assess treatment response after neoadjuvant therapy and guiding clinical decision-making and surgical planning. However, the interpretation of these images, particularly in the context of ICI, can be challenging due to the unique patterns of response, such as immune-related pseudoprogression, the appearance of tumor growth without true progressive disease or nodal immune flare (NIF), and requires careful consideration to avoid premature treatment discontinuation. NIF, defined as the apparent radiographic progression of lymph nodes without pathological evidence of tumor, was first noted in the NEOSTAR phase 2 study of nivolumab or nivolumab plus ipilimumab for resectable NSCLC and occurred in 16% of patients following neoadjuvant ICIs, leading to additional presurgical workup in 9% of patients.⁴⁰ Three patients underwent additional extensive mediastinal nodal sampling with one patient undergoing fine-needle aspiration of a newly abnormal left submandibular node, all revealing the absence of cancer and evidence of noncaseating granulomas. In response to this concern, additional measurement criteria such as irRECIST, iRECIST, and imRECIST have been proposed to measure clinical response in ICI therapy, however, due to the inherent increased complexity and suspected marginal benefit, they have not been adopted for neoadjuvant studies.⁴¹ Integrating emerging biomarkers, such as ctDNA analysis, may complement imaging modalities in assessing treatment response and discern between true progression and pseudoprogression or NIF.

ctDNA AS A BIOMARKER AND FUTURE PERSPECTIVES

An additional benefit of administering immunotherapy in combination with chemotherapy in the neoadjuvant setting is that it may enhance tumor neoantigen production, thus having a synergistic

effect to treat distant micro-metastatic disease at the earliest available time, preventing tumor migration and disease recurrence. These neoantigens released by neoadjuvant therapy may provoke a prolonged T-cell immune response that may remain functionally sustainable even after resection. Development of noninvasive liquid biopsy methods based on analyses of ctDNA provides an opportunity for monitoring of disease activity through sensitive and specific direct detection of ctDNA. Next-generation sequencing (NGS) technologies, such as whole genome sequencing or whole exome sequencing together with advanced bioinformatics workflows, have brought ctDNA-based assays to the forefront of genotyping in a variety of cancer types. However, most studies have been applied to patients with advanced cancers or have used tumor derived tissue sequencing to guide mutational analyses in plasma.^{42–51} ctDNA analysis methodologies can also be limited by low sensitivity and specificity, with many patients lacking detectable mutations or genetic alterations, and may be confounded by clonal hematopoiesis. Monitoring of ctDNA changes in early-stage cancers is challenging due to their exceptionally low fraction, usually less than 0.01% of total cell-free DNA concentration.⁵² ctDNA clearance is defined as lack of detectable mutation on NGS panels with a predefined sequence depth, breadth and detection limit (usually 0.2%). Current ongoing trials (NCT04358562 and NCT04093167) provide examples of a treatment escalation based on a molecular response adaptation approach in the setting of ctDNA clearance, however, they are primarily in the metastatic setting.^{42–46} Evaluation of ctDNA clearance kinetics in the perioperative setting (neoadjuvant and adjuvant) can act as a dynamic response biomarker and can be predictive of pCR in several solid tumors.^{53,54,55} It is imperative that prospective interventional studies incorporate the utility of ctDNA clearance kinetics as end points to determine surrogacy and the predictive capability of these measures for survival outcomes, DFS, and ultimately OS.

Current techniques to improve ctDNA detection and sensitivity rely primarily on increasing depth of sequencing combined with several error correction methodologies. However, this methodology can be error prone because with deeper sequencing comes the risk of false-positive results from detection of mutations in noncancerous cells, clonal expansions in normal epithelia, or clonal hematopoiesis.⁵⁶ Clonal hematopoiesis may be differentiated by comparing variants in ctDNA to matched tumor tissue detected by NGS. However, improved methodologies are needed to overcome these hurdles to become clinically relevant and routinely used. Detection methods that are primarily focused on genomic alterations, and mutations do not take advantage of the potential differences in chromatin organization, fragment sizes, or fragmentation patterns (fragmentomics) of ctDNA or proteomics.

Most of the phase 3 neoadjuvant studies have incorporated 12 months of adjuvant ICI, yet there is limited evidence to support such an approach, even in the setting of a pCR at resection.⁵⁷ Improved patient selection by potentially stratifying postoperative patients based on minimal residual disease (MRD) status, as detected by ctDNA, will allow us to concentrate on patients at higher risk of

recurrence, and potentially limit adjuvant treatment in those with excellent responses to surgery. This stratification strategy will facilitate the avoidance of therapies that risk potentially lifelong toxicities (particularly endocrinopathies), and optimize therapeutic outcomes to prolong the time to recurrence (DFS) in high-risk patients. The prognostic and predictive biomarker capabilities of both MRD detection and PD-L1 expression have been successfully demonstrated as a feasible stratification strategy. The presence of MRD (ctDNA+) before adjuvant therapy appeared to be associated with poorer outcomes and lower DFS for both PD-L1 expressors (>1%) and nonexpressors (<1%), and the use of adjuvant atezolizumab in the PD-L1 expressors had a similar magnitude of benefit for both the MRD-positive and MRD-negative subgroups (ctDNA+: HR, 0.54; 95% CI, 0.31–0.93; ctDNA–: HR, 0.57; 95% CI, 0.36–0.90).⁵⁸ Patients who were nonexpressors and were MRD-negative did not appear to derive significant additional benefit from the adjuvant therapy with ICI (HR, 0.95; 95% CI, 0.60–1.50).

As illustrated above, identifying MRD through detection of ctDNA in the postoperative setting could potentially predict disease relapse. MRD molecular assessment could facilitate more selective adjuvant therapy for MRD-positive patients, allowing for treatment intensification for this potentially biologically distinct disease while minimizing overtreatment of MRD negative patients. Following this molecular response adaptation design has the potential to delineate treatment intensification in nonresponders or de-escalation and even cessation of therapy in the setting of ctDNA clearance.

NEOADJUVANT AND PERIOPERATIVE IMMUNOTHERAPY CLINICAL TRIALS IN NSCLC

Early phase studies primarily focused on feasibility and safety of neoadjuvant immunotherapy; however, in the past 2 years, five large phase 3 trials; CheckMate 816, KEYNOTE-671, NEOTORCH, AEGEAN, and CheckMate-77T have reported efficacy results.^{53,59–62} Table 1 summarizes the particular characteristics of the selected literature from the published phase 1, phase 2, and phase 3 studies. Table 2 reviews the pertinent efficacy and safety end points. The tables are divided into those who received single agent immunotherapy, dual combination immunotherapy or chemotherapy and immunotherapy. Retrospective cohort studies were excluded due to a lack of clarity regarding efficacy or safety data and potential overlap with clinical trial cohorts as well as studies including radiotherapy. Neoadjuvant chemotherapy regimens typically included a platinum agent—cisplatin, carboplatin, or nedaplatin in combination with a histologically based partner: either taxane (docetaxel, paclitaxel, or nab-paclitaxel) or gemcitabine if squamous cell carcinoma, or pemetrexed if nonsquamous. All included patients underwent a minimum of one cycle and up to a maximum of four neoadjuvant cycles. Surgical resection typically took place 4–6 weeks post cessation of treatment. Patients with oncogenic drivers have typically been excluded from neoadjuvant ICI trials, with the exception of

TABLE 1 Selected neoadjuvant trials administering immunotherapy or chemo-immunotherapy for patients with resectable stage 2A to 3B NSCLC (8th edition of the AJCC Cancer Staging Manual).

Source	Type of publication	Study type	Study name/ NCT no.	Design	Sample size (N)	Follow-up (months)	Treatment	No. of neoadjuvant cycles	Primary end point
Single agent Immunotherapy									
Forde 2018 ⁶³	Article	Phase 1/2	CheckMate 159 NCT02259621	Single-arm	21	12	Nivolumab	2	Feasibility, safety
Gao 2020 ⁶⁴	Article	Phase 1b	ChiCTR-OIC-17013726	Single-arm	40	3.2	Sintilimab	2	safety
Eichhorn 2021 ⁶⁵	Article	Phase 2	NEOMUN NCT03197467	Single-arm	15	NR	Pembrolizumab	2	Feasibility, safety
Chaft 2021 ⁶⁶ and Lee 2021 ⁶⁷	Article	Phase 2	LCMC3 NCT02927301	Single-arm	181	NR	Atezolizumab	2	MPR
Besse 2020 ⁶⁸	Abstract	Phase 2	PRINCEPS NCT02994576	Single-arm	30	NR	Atezolizumab	1	Safety
Tong 2022 ⁶⁹	Article	Phase 2	TOP1501 NCT02818920	Single-arm	30	11	Pembrolizumab	2	Feasibility, safety
Bar 2019 ⁷⁰	Abstract	Phase 1	MK3475-223 NCT02938624	Single-arm	10	NR	Pembrolizumab	2	Feasibility, safety
Wislez 2020 ⁷¹ and Wislez 2022 ⁷²	Article	Phase 2	IONESCO NCT03030131	Single-arm	46	NR	Durvalumab	3	Feasibility
Dual agent-immunotherapy									
Cascone 2021 ³⁰	Article	Phase 2	NEOSTAR NCT03158129	Two-arm randomized	44	22.2	Nivolumab Nivolumab/ ipilimumab	3 3/1	MPR
Reuss 2020 ³³	Article	Phase 1b/2	NCT02259621	Single-arm	9	NR	Nivolumab/ ipilimumab	3/1	Feasibility, safety
Cascone 2023 ⁷³	Article	Phase 2	NEOCOAST NCT03794544	Four-arm umbrella study	83	NR	Durvalumab- oleclumab Durvalumab- monalizumab Durvalumab- danvatirsen Durvalumab	1	MPR
Aigner 2023 ⁷⁴	Abstract	Phase 2	NeoPredict-Lung NCT04205552	Two-arm randomized	60	NR	Nivolumab/ relatlimab	2	Feasibility
Chemoimmunotherapy									
Forde 2022 ⁵³	Article	Phase 3	CheckMate-816 NCT02998528	Two-arm randomized	358	21	Chemo/ nivolumab Chemo	3	pCR, EFS
Wakelee 2023 ⁵⁹	Article	Phase 3	KEYNOTE-671 NCT03425643	Two-arm randomized	797	25.2	Chemo/ pembrolizumab Chemo	4	EFS, OS
Heymach 2023 ⁶¹	Article	Phase 3	AEGEAN NCT03800134	Two-ARM Randomized	740	11.7	Chemo/ durvalumab Chemo	4	EFS, pCR
Lu 2023 ⁶⁰	Article	Phase 3	NEOTORCH NCT04158440	Two-arm randomized	404	18.3	Chemo/ toripalimab Chemo	4	EFS, MPR
Cascone 2023 ⁶²	Article	Phase 3	CheckMate 77T NCT04025879	Two-arm randomized	461	15.7	Chemo/ nivolumab Chemo	4	EFS

(Continues)

TABLE 1 (Continued)

Source	Type of publication	Study type	Study name/ NCT no.	Design	Sample size (N)	Follow-up (months)	Treatment	No. of neoadjuvant cycles	Primary end point
Zhang 2022 ⁷⁵	Abstract	Phase 2	ChiCTR2100044645	Single-arm	26	NR	Chemo/ camrelizumab	2–4	MPR
Lei 2023 ⁷⁶	Article	Phase 2	TD-FOREKNOW NCT04338620	Two-arm randomized	88	14.1	Chemo/ camrelizumab Chemo	3	pCR
Wu 2022 ⁷⁷	Abstract	Phase 1b	NCT04316364	Single-arm	37	NR	Chemo/SHR- 1316	3	MPR
Qiu 2022 ⁷⁸ and Shao 2023 ⁷⁹	Article	Phase 2	neoSCORE NCT04459611	Two-arm randomized	60	NR	Chemo/ sintilimab	2 or 3	MPR
Zhang 2022 ⁸⁰	Article	Phase 2	Lung-Mate001 ChiCTR1900023758	Single-arm	50	13.6	Chemo/ sintilimab	2–4	safety, MPR
Zhou 2021 ⁵⁸ and Rothschild 2021 ⁸¹	Article	Phase 2	SAKK 16/14 NCT02572843	Single-arm	67	28.6	Chemo followed by durvalumab	2	1-year EFS
Zhao 2021 ⁸²	Article	Phase 2	NeoTPD01 NCT04304248	Single-arm	33	10.13	Chemo/ toripalimab	3	MPR
Provencio 2020 ⁸³	Article	Phase 2	NADIM NCT03081689	Single-arm	46	24	Chemo/ nivolumab	3	2-year PFS
Provencio 2023 ⁸⁴	Article	Phase 2	NADIM II NCT03838159	Two-arm randomized	87	26.1	Chemo/ nivolumab Chemo	3	pCR
Shen 2021 ⁸⁵	Article	Phase 2	IRB-2021-16	Single-arm	37	7	Chemo/ pembrolizumab	2	pCR
Shu 2020 ⁸⁶	Article	Phase 2	NCT02716038	Single-arm	30	12.9	Chemo/ atezolizumab	4	MPR
Tfayli 2020 ⁸⁷	Article	Phase 2	NCT03480230	Single-arm	15	10	Chemo/ avelumab	4	ORR
Yang 2018 ⁸⁸	Article	Phase 2	TOP1201 NCT01820754	Single-arm	24	24	Chemo/ ipilimumab	2	Feasibility, safety

Abbreviations: AJCC, American Joint Committee on Cancer; chemo, chemotherapy platinum doublet; ctDNA, circulating tumor DNA; EFS, event-free survival; MPR, major pathological response; NCT, National Clinical Trial; NR, not recorded; NSCLC, non-small cell lung cancer; ORR, overall response rate; OS, overall survival; pCR, pathological complete remission; PFS, progression-free survival.

KEYNOTE-671, which randomized patients to either pembrolizumab and chemotherapy or chemotherapy, and included patients with *EGFR* or *ALK* alterations as molecular testing was not mandatory. Before initiating neoadjuvant treatment, oncogene biomarker testing which typically involves molecular diagnostics for *EGFR* mutations and *ALK* rearrangements, should also include testing for *ROS1* and *RET* fusions, *BRAF* and *MET* exon 14 skipping mutations. However, the process faces logistical and technical challenges such as insufficient or poor-quality biopsy sampling that must be addressed to maximize the utility of ICI therapy. Establishing standardized protocols for tissue handling and rapid reflex biomarker testing with *ALK* immunohistochemistry (IHC) or single-gene PCR with confirmatory NGS can improve consistency and reliability of results and turnaround times. Additionally, ctDNA analysis is rapidly establishing a role in detecting actionable mutations particularly when tissue is insufficient due to its short turnaround time.⁸⁹ To report the

outcomes uniformly, we used an ITT model of the full available analysis set with the pCR rate and MPR rate defined as the number of patients with pCR and MPR divided by total evaluable patients, those that were deemed eligible, enrolled, and received at least one dose of planned therapy. The “surgical resection rate” denotes patients whom curative-intent thoracic surgery was attempted. The “R0 resection rate” is the number of R0 resections divided by the number of patients enrolled. All patients who signed consent and began protocol treatment were considered evaluable. Reporting in a uniform and reproducible manner allows for valid cross-study comparisons. In earlier phase 1 and 2 studies, many of the reported R0 resection rates, pCR rates, and MPR rates use a modified ITT or per-protocol definitions reporting outcomes from only patients who were resected, which can lead to selection bias and can potentially over inflate the efficacy of the data, particularly if resection rates were lower than anticipated.

TABLE 2 Surgical feasibility, safety, and efficacy outcomes for selected trials.

Source	Study name/ NCT no.	Sample size (N)	Study arms	Surgical resection rate, No. (%)	RO resection rate, No. (%)	Surgical delay rate, No. (%)	Incidence of TRAE, %	Grade ≥ 3 irAEs or TRAE, %	MPR rates, %	pCR rates, %
Single agent immunotherapy										
Forde 2018 ⁶³	CheckMate 159 NCT02259621	21		20 (95)	20 (95)	0 (0)	23	5	43 (9/21)	10 (2/21)
Gao 2020 ⁶⁴	ChiCTR-OIC- 17013726	40		37 (93)	36 (90)	2 (5)	52.5	12.5	37.5 (15/ 40)	15 (6/40)
Eichhorn 2021 ⁶⁵	NEOMUN NCT03197467	15		15 (100)	15 (100)	0(0)	33	20	27 (4/15)	13 (2/15)
Chaft 2021 ⁶⁶ and Lee 2021 ⁶⁷	LCMC3 NCT02927301	181		159 (88)	145 (80)	19 (12)	61	10	16 (29/ 181)	4.4 (8/ 181)
Besse 2020 ⁶⁸	PRINCEPS NCT02994576	30		30 (100)	29 (97)	0 (0)	3	0	0 (0/ 30)	0 (0/ 30)
Tong 2022 ⁶⁹	TOP1501 NCT02818920	30		25 (83)	22 (73)	1 (4)	83	3	23 (7/30)	10 (3/30)
Bar 2019 ⁷⁰	MK3475-223 NCT02938624	10		10 (100)	NR	0 (0)	NR	NR	40 (4/10)	NR
Wislez 2020 ⁷¹ and Wislez 2022 ⁷²	IONESCO NCT03030131	46		43 (93)	41 (89)	0 (0)	34.7	0	17 (8/46)	7 (3/46)
Dual agent-immunotherapy										
Cascone 2021 ³⁰	NEOSTAR NCT03158129	23 21	Nivolumab Nivolumab/ Ipilimumab	22 (96) 17 (81)	22 (96) 17 (81)	8 (22)	NR	13 10	22 (5/23) 38 (8/21)	9 (2/23) 29 (6/21)
Reuss 2020 ³³	NCT02259621	9		6(66)	6 (66)	0(0)	67	33	22 (2/9)	22 (2/9)
Cascone 2023 ⁷³	NEOCOAST NCT03794544	21 20 16 26	Durvalumab- oleclumab Durvalumab- monalizumab Durvalumab- danvatirsen Durvalumab	18 (86) 18 (90) 16(100) 24 (92.3)	17 (81) 18 (90) 15(93.8) 22 (84.6)	4 (5)	57.1 50 43.8 34.6	4.8 0 6.3 0	19 (4/21) 30 (6/20) 31.3 (5/16) 11.2 (3/26)	9.5 (2/21) 10 (2/20) 12.5 (2/16) 3.8 (1/26)
Aigner 2023 ⁷⁴	NeoPredict-Lung NCT04205552	30 30	Nivolumab Nivolumab/ relatlimab	60 (100)	57 (95)	0 (0)	NR	NR	27 (8/30) 30 (9/30)	13.3 (4/30) 16.7 (5/30)
Chemoimmunotherapy										
Forde 2022 ⁵³	CheckMate-816 NCT02998528	179 179	Chemo/ nivolumab Chemo	149 (83) 135 (75)	124 (69.2) 105 (58.6)	31 (21) 24 (18)	82 89	33.5 36.9	36.9 (66/ 179) 8.9 (16/ 179)	24 (43/ 179) 2.2 (4/ 179)

(Continues)

TABLE 2 (Continued)

Source	Study name/ NCT no.	Sample size (N)	Study arms	Surgical resection rate, No. (%)	RO resection rate, No. (%)	Surgical delay rate, No. (%)	Incidence of TRAE, %	Grade ≥ 3 irAEs or TRAE, %	MPR rates, %	pCR rates, %
Wakelee 2023 ⁵⁹	KEYNOTE-671 NCT03425643	397 400	Chemo/ pembrolizumab Chemo	325 (82) 317 (79)	299 (75.3) 267 (66.7)	NR	97 95	44.9 37.3	30.2 (120/ 397) 11 (44/ 400)	18.1 (72/ 397) 4 (16/ 400)
Heymach 2023 ⁶¹	AEGEAN NCT03800134	366 374	Chemo/ durvalumab Chemo	295 (81) 302 (81)	269 (73) 262 (70)	58 (14.5) ^a 67 (16.8) ^a	96.5 ^a 96.5 ^a	42.4 ^a 43.2 ^a	33.3 (122/ 366) 12.3 (46/ 374)	17.2 (63/ 366) 4.3 (16/ 374)
Lu 2023 ⁶⁰	NEOTORCH NCT04158440	202 202	Chemo/ toripalimab Chemo	NR	NR	NR	99.5 98.5	63.4 54.0	48.5 (98/ 202) 8.4 (17/ 202)	24.8 (50/ 202) 1 (2/ 202)
Cascone 2023 ⁶²	CheckMate-77T	229 232	Chemo/ nivolumab Chemo	179 (78) 179 (77)	159 (69) 161 (69)	NR NR	NR NR	32 25	35.4 (81/ 229) 12.1 (28/ 232)	25.3 (58/ 229) 4.7 (11/ 232)
Zhang 2022 ⁷⁵	ChiCTR2100044645	26		17 (65)	NR	NR	38.4	7.6	38.5 (10/ 26)	19.2 (5/26)
Lei 2023 ⁷⁶	TD-FOREKNOW NCT04338620	43 45	Chemo/ camrelizumab Chemo	40 (93) 42 (93)	37 (86) 36 (80)	5 (12.5) 1 (2.4)	95.3 88.9%	25.6 11.1	65.1 (28/ 43) 15.6 (7/45)	32.6 (14/ 43) 8.9 (4/45)
Wu 2022 ⁷⁷	NCT04316364	37		34 (92)	NR	NR	100	78.4	51 (19/ 37)	30 (11/ 37)
Qiu 2022 ⁷⁸ and Shao 2023 ⁷⁹	neoSCORE NCT04459611	29 31	2 cycles Chemo/ sintilimab 3 cycles Chemo/ sintilimab	26 (90) 29 (94)	26 (90) 29 (94)	NR	NR	31.0 29	24 (7/29) 39 (12/ 31)	17 (5/29) 23 (7/31)
Zhang 2022 ⁸⁰	Lung-Mate001 ChiCTR1900023758	50		30 (60)	30 (60)	0 (0)	95	8	26 (13/ 50)	12 (6/50)
Rothschild 2021 ⁸¹	SAKK 16/14 NCT02572843	67		55 (82)	51 (76)	NR	100	88	51 (34/ 67)	15 (10/ 67)
Zhao 2021 ⁸²	NeoTPD01 NCT04304248	33		30 (91.9)	29 (88)	0 (0)	91	18.1	60.6 (20/ 33)	45.4 (15/ 33)
Provencio 2020 ⁸³	NADIM NCT03081689	46		41 (89)	41 (89)	0 (0)	93	34	74 (34/ 46)	56.5 (26/ 46)

TABLE 2 (Continued)

Source	Study name/ NCT no.	Sample size (N)	Study arms	Surgical resection rate, No. (%)	RO resection rate, No. (%)	Surgical delay rate, No. (%)	Incidence of TRAE, %	Grade ≥ 3 irAEs or TRAE, %	MPR rates, %	pCR rates, %
Provencio 2023 ⁸⁴	NADIM II NCT03838159	57 29	Chemo/ nivolumab Chemo	53 (93) 20 (69)	50 (88) 17 (59)	0 (0)	88 90	24 10	53 (30/ 57) 14 (4/29)	37 (21/ 57) 7 (2/29)
Shen 2021 ⁸⁵	IRB-2021-16	37		37 (100)	37 (100)	0 (0)	70.3	10.8	65 (24/ 37)	46 (17/ 37)
Shu 2020 ⁸⁶	NCT02716038	30		29 (97)	26 (87)	0 (0)	87	50	57 (17/ 30)	33 (10/ 30)
Tfayli 2020 ⁸⁷	NCT03480230	15		11(73)	NR	0 (0)	NR	27	20 (3/15)	7 (1/15)
Yang 2018 ⁸⁸	TOP1201 (NCT01820754)	24		13 (54)	13 (54)	2 (8)	100	46	NR	8 (2/24)

Abbreviations: chemo, chemotherapy platinum doublet; irAE, immune-related adverse events; MPR, major pathological response; NCT, National Clinical Trial; NR, not recorded; NSCLC, non-small cell lung cancer; pCR, pathological complete remission; PFS, progression-free survival; RO, complete resection with clear margins; TRAE, treatment-related adverse events.

^aThe AEGEAN study included EGFR/ALK patients in the safety/delay data and excluded them in the surgical and efficacy data.

EFFICACY OF NEOADJUVANT IMMUNOTHERAPY IN NSCLC

Neoadjuvant immunotherapy, either as single agent or doublet, or in combination with chemotherapy, has shown extremely encouraging efficacy outcomes in individuals diagnosed with NSCLC. A total of 29 of 30 selected studies (Table 2) reported efficacy data in the form of pCR and MPR rates. Efficacy and toxicity are directly proportional when treatment is escalated. The pooled median pCR and MPR rates for single agent PD-(L)1 were 9.5% (95% CI, 6.37%–12.63%) and 23% (95% CI, 15.35%–30.65%), respectively, with pCR ranging from 0% to 15% and MPR ranging from 0% to 43%. CheckMate 159, the seminal pilot neoadjuvant immunotherapy trial using nivolumab for two doses met its primary end point of safety with any grade immune-related adverse events (irAE) reported in 23% and serious (grade 3–5) irAE in 5%. pCR and MPR rates were 10% (2 of 21) and 43% (9 of 21), respectively.⁶³ The updated 5-year EFS was 60%.⁹⁰ The pooled pCR and MPR rates for immunotherapy doublet were 14.6% (95% CI, 8.47%–20.73%) and 30% (95% CI, 24.51%–35.49%), respectively, with pCR ranging from 10% to 29% and MPR ranging from 30% to 38%. The phase 2 NEOSTAR study randomized eligible patients to either three cycles of nivolumab or ipilimumab for one cycle and three cycles of nivolumab. It had increased pathological regression as anticipated for the combination arm with pCR and MPR rates of 29% (6 of 21) and 38% (8 of 21), compared to 9% (2 of 23) and 22% (5 of 23) for nivolumab.³⁰ The ipilimumab and nivolumab arm of the CheckMate 816 was closed early but demonstrated pCR rates of 20% (95% CI, 13.4%–29.0%) and EFS was 54.8 months with a 3-year EFS rates of 56%.⁹¹ The NEOCOAST platform trial combined

durvalumab with other checkpoint inhibitors including oleclumab (anti-CD73), monalizumab (anti-NKG2A), and danvatrisen (anti-STAT3 antisense oligonucleotide). pCR (9.5%, 10%, and 12.5%, respectively) and MPR (19.0%, 30%, and 31.3%, respectively) rates were higher in the combination arms compared to durvalumab (MPR, 11%; pCR, 4%).⁷³

The pooled pCR and MPR rates for combined chemo-immunotherapy were 24% (95% CI, 18%–30%) and 43.75% (95% CI, 36.49%–51.01%), respectively, with pCR ranging from 7% to 56.5% and MPR ranging from 20% to 65.1%. Most chemoimmunotherapy trials delivered concurrent chemotherapy and ICI. Notably the phase 2 SAKK 16/14 study treating high risk stage IIIA NSCLC was the only trial to deliver sequential therapy with two doses of durvalumab after neoadjuvant doublet chemotherapy before resection. The pCR and MPR rate was 15% (10 of 67) and 51% (34 of 67) respectively, and the 1-year event-free survival (EFS) rate was 73% (two-sided 90% CI, 63%–82%).⁸¹ The phase 2 NADIM single arm study that delivered three cycles of nivolumab plus platinum-based chemotherapy enrolled only stage IIIA patients and had higher than anticipated pathological regression with pCR and MPR rates of 56.5% (26 of 46) and 74% (34 of 46) with a corresponding PFS at 24 months of 77.1% (95% CI, 59.9%–87.7%).⁸³ Increased levels of PD-L1 or tumor mutational burden (TMB) were not found to be contributing factors to these findings, although ctDNA clearance kinetics and dynamics was predictive of response.⁹² The phase 2 NADIM II study enrolled both stage IIIA and IIIB patients and also delivered three cycles of nivolumab plus platinum-based chemotherapy and its results were marginally lower with the combination arm demonstrating pCR and MPR rates of 37% (21 of 57) and 53% (30 of 57) with a corresponding

PFS at 24 months of 67.2% (95% CI, 55.8%–81.0%). Notably, the patients who achieved a pCR were disease-free and alive at the time of data cutoff. The predictive value of PD-L1 TPS for pCR (area under the curve) was 0.728 (95% CI, 0.58–0.87).⁸⁴ A recent meta-analysis of PD-L1 expression and TMB as pathological response biomarkers examined these findings further across many of the studies outlined in Table 2. It found that high TMB was associated with MPR (odds ratio [OR], 3.40; 95% CI, 1.33–8.70; $p = .0109$) and pCR (OR, 1.98; 95% CI, 1.08–3.63; $p = .0265$). It also found that PD-L1 TPS expression greater than 1% was associated with higher rate of MPR (OR, 2.62; 95% CI, 1.51–4.56; $p = .0006$) and pCR (OR, 2.94; 95% CI, 1.69–5.09; $p = <.0001$) but the probability can vary between drug (PD-1 vs. PD-L1 inhibitor) and testing methodology (IHC [28–8] vs. IHC [22C3] vs. IHC [SP263]).⁹³ The added benefit of PD-(L)1 to chemotherapy was confirmed by the results of the phase 3 studies CheckMate 816, KEYNOTE-671, AEGEAN, NEOTORCH, and Checkmate 77T. CheckMate 816, which delivered only three doses of neoadjuvant treatment, either platinum-based chemotherapy or nivolumab plus platinum-based chemotherapy, met both its coprimary end points by demonstrating improvements in pCR and EFS.⁵³ pCR rates improved from 2.2% (95% CI, 0.6%–5.6%) in the chemotherapy arm, similar to historical results, to 24% (95% CI, 18.0%–31.0%) in the immunochemotherapy arm. EFS was significantly longer at 31.6 months (95% CI, 30.2–not reached (NR)) with the addition of nivolumab compared to 20.8 months (95% CI, 14.0–26.7) for the chemotherapy arm (HR, 0.63; $p = .005$). This corresponded to PFS at 24 months of 63.8%. Similar to NADIM II, the magnitude of benefit was increased for patients with a tumor PD-L1 TPS expression greater or equal to 1% more than in those with a TPS score less than 1%.⁵³ The remaining phase 3 studies delivered perioperative immunotherapy, both four cycles of neoadjuvant and 12 months of adjuvant therapy. KEYNOTE-671, which delivered four doses of neoadjuvant treatment, either platinum-based chemotherapy or pembrolizumab plus platinum-based chemotherapy, followed by 1 year of adjuvant placebo or pembrolizumab, was deemed positive after it met its first coprimary end point by significantly improving EFS.⁵⁹ At a median follow-up of 36.6 months the EFS was 47.2 (95% CI, 32.9–NR) with the addition of pembrolizumab compared to 18.3 months (95% CI, 14.8–22.1) for the chemotherapy arm (HR, 0.59; $p = <.00001$). This corresponded to EFS at 24 months of 62.4%. The OS was not reached (95% CI, NR–NR) with the addition of pembrolizumab compared to 52.4 months (95% CI, 45.7–NR) for the chemotherapy arm (HR, 0.72; $p = .00517$).⁹⁴ pCR rates improved from 4% (95% CI, 2.3%–6.4%) in the chemotherapy arm, to 18.1% (95% CI, 14.5%–22.3%).⁵⁹ AEGEAN, which delivered four doses of neoadjuvant treatment, either platinum-based chemotherapy or durvalumab plus platinum-based chemotherapy, followed by 1 year of adjuvant placebo or durvalumab, met both its coprimary end points by demonstrating improvements in pCR and EFS.⁶¹ It is worthwhile noting the protocol was amended during enrollment to exclude oncogenic driven tumors EGFR mutations and ALK translocations. pCR rates improved from 4.3% (95% CI, 2.5%–6.9%) in the chemotherapy arm, to 17.2% (95% CI, 13.5%–21.5%). The EFS was

not reached (95% CI, 31.9–NR) with the addition of durvalumab compared to 25.9 months (95% CI, 18.9–NR) for the chemotherapy arm (HR, 0.68; $p = <.004$). This corresponded to EFS at 24 months of 63.3% in the modified ITT population.⁶¹ Both NEOTORCH, which delivered four doses of neoadjuvant treatment (either platinum-based chemotherapy or toripalimab plus platinum-based chemotherapy) followed by 1 year of adjuvant placebo or toripalimab and CheckMate 77T, which delivered four doses of neoadjuvant treatment (either platinum-based chemotherapy or nivolumab plus platinum-based chemotherapy) followed by 1 year of adjuvant placebo or nivolumab were presented at American Society of Clinical Oncology plenary session 2023 and the Presidential Symposium at the European Society for Medical Oncology Congress 2023, respectively, and demonstrated positive results with improvements in EFS with the addition of toripalimab (HR, 0.40; 95% CI, 0.28–0.57; $p = <.0001$) and nivolumab (HR, 0.58, 97.36% CI, 0.42–0.81; $p = 0.00025$).^{60,62} A recent exploratory, retrospective patient-level data analysis comparing perioperative Nivolumab (CheckMate 77T) to neoadjuvant Nivolumab alone (CheckMate 816) was presented at World Conference on Lung Cancer 2024.⁵⁷ Given the nature of this cross-trial comparison propensity score weighting was used, both average treatment effect for the treated (ATT) and average treatment effect (ATE), to adjust for imbalances in baseline demographics and disease characteristic variables. Perioperative nivolumab improved EFS over neoadjuvant nivolumab in the weighted and unweighted analyses (ATE: HR, 0.61; 95% CI, 0.39–0.97; ATT: HR, 0.56; 95% CI, 0.35–0.90; unweighted: HR, 0.59; 95% CI, 0.38–0.92). There was also a trend toward improved EFS for patients who achieved a pCR and those that did not (pCR: HR, 0.58; 95% CI, 0.14–2.40; no pCR: HR, 0.65; 95% CI, 0.40–1.06), and it appears the magnitude of benefit of the adjuvant nivolumab component is driven by those with TPS PD-L1 less than 1% (PD-L1 <1%: HR, 0.51; 95% CI, 0.28–0.93; PD-L1 >1%: HR, 0.86; 95% CI, 0.44–1.70) although this analysis was unweighted and limited by patient numbers.

SAFETY OF NEOADJUVANT IMMUNOTHERAPY IN NSCLC

Neoadjuvant treatment, regardless of the various approaches used, is generally well-tolerated in patients with potentially resectable NSCLC. As described in Table 2 the safety outcomes, including any reported treatment-related adverse events (TRAE) or serious treatment-related adverse events (TRAE grade 3–5), were reported in 28 trials. Serious TRAE were consistently low for single agent or combination immunotherapy but toxicity rates increased as expected with the addition of chemotherapy. The incidence of any grade and grade 3–5 irAE for single agent PD-(L)1 were 43.6% (95% CI, 25.4%–61.8%) and 10% (95% CI, 5%–15%) and were 50% (95% CI, 40%–60%) and 6% (95% CI, 0%–15.8%) for immunotherapy doublet. In the NEOSTAR study, serious irAEs were similar between the arms (nivolumab alone, 13% [3 of 23] and nivolumab + ipilimumab, 10% [2 of 21]).³⁰ However, the combination arm of CheckMate 159, which

included three cycles of nivolumab and one cycle of ipilimumab, a similar dosing and schedule to NEOSTAR, was terminated early due to higher than anticipated serious toxicity (3 of 9, 33%).³³ The incidence of any grade and grade 3–5 TRAE for chemoimmunotherapy were 95% (95% CI, 87%–100%) and 32% (95% CI, 22%–42%) and were comparable to chemotherapy alone (93% [95% CI, 89.7%–96.3%] and 36.9% [95% CI, 24.8%–49%]). The rate of grade 3 or worse TRAE was not significantly different between the experimental and control arms for each of the phase 3 trials. CheckMate 816; 33.5% in the nivolumab arm and 36.9% in the chemotherapy alone arm. KEYNOTE-671; 44.9% in the pembrolizumab arm and 37.3% in the chemotherapy alone arm. AEGEAN; 42.4% in the durvalumab arm and 43.2% in the chemotherapy alone arm. In the phase 2 NADIM study, 34% of patients developed TRAE of grade 3 or worse.⁸³ None of the adverse events led to treatment discontinuation, dose reduction, surgery delay, or death. In contrast, the IONESCO phase 2 trial, which delivered three cycles of single agent durvalumab, was halted prematurely due to excessive 90-day postoperative mortality; although the deaths were not directly linked to the preoperative durvalumab therapy, caution should be exercised with any curative intent therapies.⁷¹

Other key indicators for assessing safety of neoadjuvant treatment are the surgical resection rate, which is defined as the percentage of patients who were deemed potential surgical candidates at the end of neoadjuvant therapy and whom curative-intent thoracic surgery was attempted, and the R0 resection rates, those patients who achieved clear margins at time of resection. These were reported in 29 and 25 trials, respectively. The pooled resection rate was 90.95% (95% CI, 86.93%–94.97%), ranging from 54% to 100%, and the R0 resection rate was similar at 87.5% (95% CI, 83.24%–91.76%), with a range from 54% to 100%, indicating that if a patient were successful in reaching surgery, they were likely to have a R0 resection. The resection and R0 resection rate for chemotherapy alone was lower at 78% (95% CI, 71.6%–84.4%) and 67.85% (95% CI, 61.49%–74.21%). CheckMate 816 demonstrated improved definitive surgical resection rates in the immunotherapy arm compared to the chemotherapy alone arm, 83.2% versus 75.4%.⁵³ KEYNOTE-671 and AEGEAN, demonstrated similar definitive surgical resection rates between arms, 82% versus 79%, and 81% versus 81%, respectively.^{59,61} The most common reason for failure to proceed with resection was disease progression or unresectable disease, patient refusal, mortality, TRAE, poor lung function, or drop out. An interesting finding from CheckMate 816 of the surgical outcomes, available in their supplemental appendix, was when patients are divided by region it appears that North America, both for the immunotherapy and chemotherapy arms, had lower R0 resection rates (64.9% vs. 50%, respectively) compared to both Europe and Asia, which coincided with a higher rate of a minimally invasive approach (MIS) which was generally not an adopted approach in Europe. A meta-analysis of the randomised controlled trials comparing open thoracotomy and MIS found that they had equivocal operative quality standards, with MIS leading to a shorter hospital stay but neither was superior in terms of surgical efficacy or complications. However, most of the

patients enrolled in these studies had very early-stage disease (stage I or II) with limited nodal involvement.⁹⁵ It would be interesting to explore this phenomenon further in the other reported phase 3 trials, but the data are not currently publicly available. The surgical delay rate, which is the ratio of patients whose surgery was delayed to all of those resected, is another key indicator for assessing safety. Only 222 of 2908 (7.6%) resected patients exhibited a surgical delay. The surgical delay rates ranged from 0% to 22%. The surgical delay rate was not reported in seven of the 30 studies. CheckMate 816 reported a 21% ($N = 31$) delay to surgery for the experimental arm and 18% ($N = 24$) for the chemotherapy arm, however, only 4% ($N = 6$) and 6.7% ($N = 9$) were a result of AEs. The surgical outcomes demonstrated similar lengths of hospitalization (10 days in both arms) and similar duration of surgery (185.0 min vs. 213.5 min). The AEGEAN trial reported a 14.5% ($N = 58$) delay to surgery for the experimental arm and 16.8% ($N = 67$) for the chemotherapy arm, however, similarly most of these delays were administrative or logistical (31 [7.7% vs. 9.3%]) rather than AE-related. These findings led to FDA and EMA approval and are recommended by National Comprehensive Cancer Network both for nivolumab and pembrolizumab plus chemotherapy in the neoadjuvant and perioperative setting for patients with potentially resectable NSCLC. PD-L1 IHC testing has been integrated into clinical practice particularly in Europe, as it determines the use of neoadjuvant nivolumab in combination with chemotherapy, which has been restricted to the subset of patients with tumor PD-L1 expression $\geq 1\%$, although it has shown benefits regardless of PD-L1 status. European approval of perioperative pembrolizumab is not dependent of PD-L1 status, and we await the outcomes of their review for perioperative toripalimab, durvalumab, and nivolumab. The above safety profile, particularly from large the randomized trials, have alleviated some of the fears regarding neoadjuvant therapy and the possibility of toxicity affecting surgical outcomes and/or causing delays in patients receiving curative surgery.

CONCLUSION AND REMAINING QUESTIONS

This review highlights the clear feasibility and safety of administering immunotherapy—whether as monotherapy, in combination with other immunotherapy agents, or combined with chemotherapy—for up to four cycles prior to surgical resection, without significant delays or adverse consequences. The literature also suggests that not only are neoadjuvant and perioperative approaches safe and feasible but have extremely efficacious results with encouraging pathological responses and improved survival outcomes. The choice of treatment paradigm and optimal sequencing between either neoadjuvant, adjuvant, or perioperative therapy remains unclear and is typically performed on a case-by-case basis after discussion at multidisciplinary meeting. It is impossible to compare neoadjuvant to adjuvant trials due to the multitude of confounding factors and differing selection criteria. Typically, neoadjuvant-treated patients are higher risk with more advanced disease at diagnosis, and therefore, direct

comparisons between outcomes are unwise. However, it may be possible in the future to compare a neoadjuvant to an adjuvant or perioperative approach prospectively as evidenced from recent studies in melanoma.^{20,96} Furthermore, clinical staging, which is used in neoadjuvant studies, is a distinct entity to pathological staging and clinically staged patients are often upstaged at surgical resection.⁹⁷ It is unlikely that we will see a study in the near future that examines and compares directly a neoadjuvant and adjuvant approach to early-stage lung cancer and they will likely remain two distinct entities. However, as we become more comfortable with a neoadjuvant approach, we will likely see its utilization grow given the additional benefits that can be gained as outlined above. However, there remain several unanswered questions that need to be addressed regarding neoadjuvant therapy.

We know that combined chemoimmunotherapy is the best treatment model in terms of efficacy, however, monotherapy is a less toxic option and demonstrates reasonable responses in select patients seen in previous phase 2 trials. We need to understand the pathophysiology of these optimal responders to allow appropriate patient selection and minimize toxicity. Another question is the appropriate duration and number of cycles of neoadjuvant treatment. Most phase 3 studies have included four cycles as standard, however, Checkmate 816 only had three. The role of adjuvant immunotherapy in this setting remains unknown and may interfere with our ability to interpret survival outcomes and the overall benefit of neoadjuvant therapy. Additionally, the role of neoadjuvant radiotherapy remains unclear, although it has been shown to augment the tumor immune milieu and demonstrate potential synergy with ICI most of the recent studies integrating concurrent or consolidation ICI have been performed in the nonoperative or locally advanced setting.^{25,98} The single arm phase 2 INCREASE study combined induction nivolumab and ipilimumab to neoadjuvant chemoradiotherapy for patients with resectable and borderline resectable disease. It demonstrated encouraging results with a pCR rate of 50% and MPR rate 63%, although with the caveat of a comparatively high rate of grade 3–4 TRAE (83%).⁹⁹ It remains unclear whether the observed effect is primarily driven by the addition of ipilimumab, radiotherapy, or a combination of both. Furthermore, if radiotherapy is the key contributor, it is uncertain whether the resulting increase translates to the same long-term advantage in preventing distant metastatic disease (beyond the radiation field) as it would if solely attributed to the systemic administration of ICI alongside chemotherapy. We await further results from ongoing phase 2 studies. Further research is needed to identify those patients who benefit most from a neoadjuvant ICI approach, but more importantly, those that do not benefit and thus may be best treated with alternative approaches.

AUTHOR CONTRIBUTIONS

Colum Dennehy: Conceptualization; methodology; writing—review and editing, and writing—original draft. **Michael R. Conroy:** Writing—review and editing. **Patrick M. Forde:** Writing—review and editing and supervision.

CONFLICT OF INTEREST STATEMENT

Michael R. Conroy reports grant and/or contract funding from the Conquer Cancer Foundation. Patrick M. Forde reports consulting fees from Amgen, AstraZeneca, Bristol-Myers Squibb, Daiichi Sankyo, F-Star, G1 Therapeutics, Genentech, Iteos, Janssen, Merck, Mirati, Novartis, Sanofi Pasteur Inc, and Surface Oncology; fees for data and safety monitoring from Flame Biosciences and Polaris; fees for other professional activities from LUNGEVITY Foundation; and research funding from the National Cancer Institute Cancer Center and Regeneron Pharmaceuticals; and research funding to Johns Hopkins University from AstraZeneca, Bristol-Myers Squibb, Novartis, Corvus, and Kyowa. The other author declares no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article because no new data were created or analyzed in this study.

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